

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-I]

TEST # 09

DATE : 01-09-2019

Test Type : Minor

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
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MINOR TEST # 09

Topics

DATE : 01 - 09 - 2019

PRE-MEDICAL ENTHUSIAST COURSE : PHASE - I – PATTERN : [NEET]

PHYSICS : MEC, Wave Optics, KTG, Thermodynamics, Thermal Physics

CHEMISTRY : Coordination Compound, Chemistry in Everyday life, Biomolecules, Polymer,
Thermodynamics, Thermochemistry, Chemical Bonding, HydrocarbonBIOLOGY : Biotechnology-I, Transportation in Plants, Mineral Nutrition, Drug And Alcohol
Abuse, Circular System

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HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

1. A pendulum clock keeps correct time at 0°C . The thermal coefficient of linear expansion of the material of the pendulum is α . If the temperature rises to $t^\circ\text{C}$ then the clock loses per day by :-

- (1) αt sec
(2) $\frac{1}{2}\alpha t$ sec
(3) $\alpha t \times 86400$ sec
(4) $\frac{1}{2}\alpha t \times 86400$ sec

2. Two rods of copper and brass having the same length and cross-section are joined end to end. The free end of the copper rod is at 0°C and of the brass rod is at 100°C in steady state. If thermal conductivity of copper is 4 times of that of brass, find the temperature at the junction of two rods:-

- (1) 80°C (2) 60°C
(3) 40°C (4) 20°C

3. A composite rod made of 3 rods of equal length and cross-section is shown in figure. The end A and B are at constant temperatures. The effective thermal conductivity of the bar is :-



- (1) $15 K/16$ (2) $6 K/13$
(3) $5 K/16$ (4) $2 K/13$

4. A body cools from 62°C to 50°C in 10 minutes and to 42°C in the next 10 minutes. The temperature of the surrounding is :-

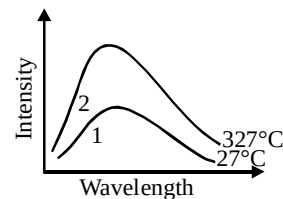
- (1) 16°C (2) 26°C
(3) 36°C (4) 21°C

5. When the temperature of a black body increases, it is observed that the wavelength corresponding to maximum energy changes from $0.26 \mu\text{m}$ to $0.13 \mu\text{m}$, the ratio of the emissive powers of the body at the respective temperatures :-

- (1) $16/1$ (2) $4/1$
(3) $1/4$ (4) $1/16$

6. The spectrum of a black body at two temperature 27°C and 327°C is shown in the figure. Let A_1 and A_2 be the areas under the two curves respectively. The

value of $\frac{A_2}{A_1}$ is :-



- (1) $1 : 16$ (2) $4 : 1$
(3) $2 : 1$ (4) $16 : 1$

7. Two stars P and Q radiate maximum energy at wavelengths $3600\text{\AA}$ and $4800\text{\AA}$ respectively. The ratio of absolute temperature of P and Q is:-

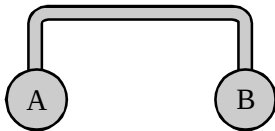
- (1) $4 : 3$ (2) $3 : 4$
(3) $256 : 81$ (4) $81 : 256$

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8. For helium gas $C_p - C_v = a$ and for oxygen gas $C_p - C_v = b$. So the relation between a and b is given by (Where C_p , C_v are gram specific heat at constant pressure and const. Volume) :-

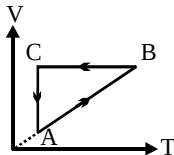
- (1) $a = 16b$ (2) $b = 16a$
(3) $a = 8b$ (4) $a = b$

9. Two spherical vessel of equal volume, are connected by a narrow tube. The apparatus contains an ideal gas at one atmosphere and 300 K. Now if one vessel is immersed in a bath of constant temperature 600 K and the other in a bath of constant temperature 300 K. Then the common pressure will be :-



- (1) 1 atm (2) $\frac{4}{5}$ atm
(3) $\frac{4}{3}$ atm (4) $\frac{3}{4}$ atm

10. A cyclic process ABCA is shown in the V-T diagram. Process on the P-V diagram is :-



- (1) (2)
(3) (4)

11. Determine the angular separation between central maximum and first order maxima of the diffraction pattern due to a single slit of width 0.25 mm when light of wavelength 5890 Å is incident on it :-

- (1) 3.534×10^{-3} rad
(2) 2.35×10^{-3} rad
(3) 1.47×10^{-4} rad
(4) 15.8×10^{-3} rad

12. A polaroid examines two adjacent polarised light beams A and B whose planes of polarisation are mutually at right angles. In one position of the polaroid, the beam B shows zero intensity. From this position, a rotation of 30° in polaroid shows the two beams of equal intensities. Find the intensity ratio I_A/I_B of the two beams.

- (1) 1 : 3 (2) $\sqrt{3} : 2$
(3) 1 : 1 (4) $1 : \sqrt{2}$

13. The central fringe of the interference pattern produced by the light of wavelength 6000 Å is found to shift to the position of 4th dark fringe after a glass sheet of refractive index 1.5 is introduced. The thickness of glass sheet would be :-

- (1) 4.8 μm (2) 4.2 μm
(3) 5.4 μm (4) 3.0 μm

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14. The main difference in the phenomenon of interference and diffraction is that :

- (1) Diffraction is due to interaction of light from the same wavefront whereas interference is the interaction of waves from two isolated sources.
- (2) Diffraction is due to interaction of light from same wavefront, whereas the interference is the interaction of two waves derived from the same source
- (3) Diffraction is due to interaction of waves derived from the same source, whereas the interference is the bending of light from the same wavefront
- (4) Diffraction is caused by the reflected waves from a source whereas interference is caused due to refraction of waves from a source

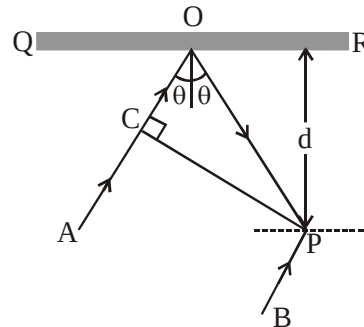
15. The equation of two light waves are $y_1 = 6 \cos \omega t$. $y_2 = 8 \cos (\omega t + \phi)$. The ratio of maximum to minimum intensities produced by the supersposition of these waves will be :

- (1) 49 : 1
- (2) 1 : 49
- (3) 1 : 7
- (4) 7 : 1

16. In Young's experiment, the distance between two slits is made three times then the fringe width will become :

- (1) 9 times
- (2) 1/9 times
- (3) 3 times
- (4) 1/3 times

17. In the given diagram, CP represents a wavefront and AO and BP, the corresponding two rays. Find the condition on θ for constructive interference at P between the ray BP and reflected ray OP



- (1) $\cos \theta = 3\lambda/2d$
- (2) $\cos \theta = \lambda/4d$
- (3) $\sec \theta - \cos \theta = \lambda/d$
- (4) $\sec \theta - \cos \theta = 4\lambda/d$

18. When light source is point source, then shape of wavefront will be :-

- (1) Cylindrical
- (2) Spherical
- (3) Plane
- (4) None of these

19. In an interference pattern, at a point we observe the 16th order maximum for $\lambda_1 = 6000 \text{ \AA}$. What order will be visible here if the source is replaced by light of wavelength $\lambda_2 = 4800 \text{ \AA}$?

- (1) 40
- (2) 20
- (3) 10
- (4) 80

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20. Interference fringes were produced in Young's double slit experiment using light of wavelength $5000 \text{ \AA}$. When a film $2.5 \times 10^{-3} \text{ cm}$ thick was placed over one of the slits, the fringe pattern shifted by a distance equal to 20 fringe width. The refractive index of the material of the film can be :
- (1) 1.25 (2) 1.33
(3) 1.5 (4) 1.4
21. Consider interference between waves from two sources of intensities I and $4I$. Find intensities at point where the phase difference is $\frac{\pi}{2}$.
- (1) I (2) $5I$
(3) $4I$ (4) $3I$
22. In Young's double slit experiment, white light source is used to obtain a white central fringe and a few coloured fringes. Now, if a filter allowing only red light is used in front of slit S_1 and another filter allowing only blue light is used in front of second slit S_2 , then :
- (1) Only red coloured fringes will be observed
(2) Only blue coloured fringes will be observed
(3) Both red coloured and blue coloured fringes will be observed
(4) Interference pattern will not form on the screen
23. In Young's double slit experiment $\frac{d}{D} = 10^{-4}$ (d = distance between slits, D = distance of screen from the slits). At a point P on the screen the resulting intensity is equal to the intensity due to individual slit I_0 . Then the distance of point P from the central maximum is : ($\lambda = 6000 \text{ \AA}$)
- (1) 2 mm (2) 1 mm
(3) 0.5 mm (4) 4 mm
24. The width of a single slit, if the first minimum is observed at an angle of 2° with a light of wavelength $6980 \text{ \AA}$, (in mm) is :
- (1) 0.2 (2) 2×10^{-5}
(3) 2×10^5 (4) 0.02
25. A beam of natural light falls on a system of 5 polaroids, which are arranged in succession such that the pass axis of each polaroid is turned through 60° with respect to the preceding one. The fraction of the incident light intensity that passes through the system is :
- (1) $\frac{1}{64}$ (2) $\frac{1}{32}$
(3) $\frac{1}{256}$ (4) $\frac{1}{512}$

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26. The magnetic field $\vec{dB}$ due to a small current element $\vec{dl}$ at a distance $\vec{r}$ and element carrying current i is,

or

Vector form of Biot-savart's law is :-

(1) $\vec{dB} = \frac{\mu_0}{4\pi} i \left(\frac{\vec{dl} \times \vec{r}}{r} \right)$

(2) $\vec{dB} = \frac{\mu_0}{4\pi} i^2 \left(\frac{\vec{dl} \times \vec{r}}{r} \right)$

(3) $\vec{dB} = \frac{\mu_0}{4\pi} i^2 \left(\frac{\vec{dl} \times \vec{r}}{r^2} \right)$

(4) $\vec{dB} = \frac{\mu_0}{4\pi} i \left(\frac{\vec{dl} \times \vec{r}}{r^3} \right)$

27. A proton (mass $= 1.67 \times 10^{-27}$ kg and charge $= 1.6 \times 10^{-19}$ C) enters perpendicular to a magnetic field of intensity 2 weber/m² with a velocity 3.4×10^7 m/sec. The acceleration of the proton should be :-
- (1) 6.5×10^{15} m/sec²
 (2) 6.5×10^{13} m/sec²
 (3) 6.5×10^{11} m/sec²
 (4) 6.5×10^9 m/sec²

28. A proton and a deuteron both having the same kinetic energy, enter perpendicularly into a uniform magnetic field B . For motion of proton and deuteron on circular path of radius R_p and R_d respectively, the correct statement is :-

- (1) $R_d = \sqrt{2}R_p$ (2) $R_d = R_p / \sqrt{2}$
 (3) $R_d = R_p$ (4) $R_d = 2R_p$

29. A charge moves in a circle perpendicular to a magnetic field. The time period of revolution is independent of :-

- (1) Magnetic field
 (2) Charge
 (3) Mass of the particle
 (4) Velocity of the particle

30. The strength of the magnetic field at a point r near a long straight current carrying wire is B . The field at a distance

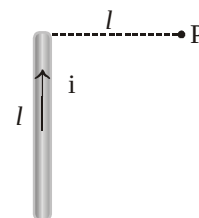
$\frac{r}{2}$ will be :-

- (1) $\frac{B}{2}$ (2) $\frac{B}{4}$ (3) $2B$ (4) $4B$

31. An arc of a circle of radius R subtends an angle $\frac{\pi}{2}$ at the centre. It carries a current i . The magnetic field at the centre will be :-

- (1) $\frac{\mu_0 i}{2R}$ (2) $\frac{\mu_0 i}{8R}$ (3) $\frac{\mu_0 i}{4R}$ (4) $\frac{2\mu_0 i}{5R}$

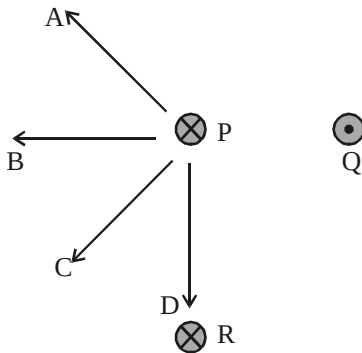
32. Figure shows a straight wire of length l current i . The magnitude of magnetic field produced by the current at point P is :-



- (1) $\frac{\sqrt{2}\mu_0 i}{\pi l}$ (2) $\frac{\mu_0 i}{4\pi l}$
 (3) $\frac{\sqrt{2}\mu_0 i}{8\pi l}$ (4) $\frac{\mu_0 i}{2\sqrt{2}\pi l}$

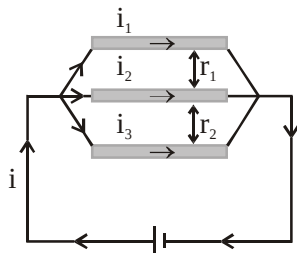
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33. In figure shows three long straight wires P, Q and R carrying currents normal to the plane of the paper. All three currents have the same magnitude. Which arrow best shows the direction of the resultant force on the wire P :-



- (1) A (2) B (3) C (4) D

34. Three long straight wires are connected parallel to each other across a battery of negligible internal resistance. The ratio of their resistances are 3 : 4 : 5. What is the ratio of distances of middle wire from the others if the net force experienced by it is zero :-



- (1) 4 : 3 (2) 3 : 1
(3) 5 : 3 (4) 2 : 3

35. The magnetic force acting on a charged particle of charge $2 \mu\text{C}$ in a magnetic field of 2 T acting in y-direction, when the particle velocity is

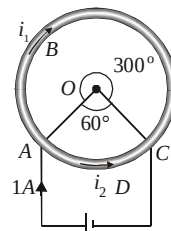
$(2\hat{i} + 3\hat{j}) \times 10^6 \text{ ms}^{-1}$ is :-

- (1) 4 N in y-direction
(2) 8 N in z-direction
(3) 8 N in y-direction
(4) 4 N in z-direction

36. A cell is connected between the points A and C of a circular conductor ABCD of centre O with angle $\text{AOC} = 60^\circ$. If B_1 and B_2 are the magnitudes of the magnetic fields at O due to the currents in ABC and ADC respectively, the ratio $\frac{B_1}{B_2}$ is :

ADC respectively, the ratio $\frac{B_1}{B_2}$ is :

- (1) 0.2
(2) 6
(3) 1
(4) 5

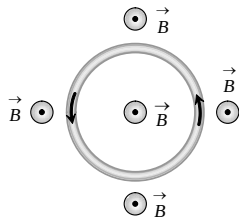


37. A wire of length L metre carrying a current of I ampere is bent in the form of a circle. Its magnitude of magnetic moment will be :

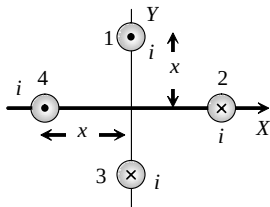
- (1) $\frac{IL}{4\pi}$ (2) $\frac{IL^2}{4\pi}$ (3) $\frac{I^2L^2}{4\pi}$ (4) $\frac{I^2L}{4\pi}$

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38. An elastic circular wire of length l carries a current I . It is placed in a uniform magnetic field (Out of paper) such that its plane is perpendicular to the direction of $\vec{B}$. The wire will experience

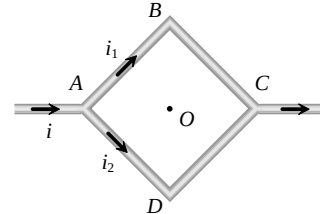


- (1) No force
(2) A stretching force
(3) A compressive force
(4) A torque
39. What will be the resultant magnetic field at origin due to four infinite length wires. If each wire produces magnetic field 'B' at origin



- (1) $4B$
(2) $\sqrt{2}B$
(3) $2\sqrt{2}B$
(4) Zero

40. Figure shows a square loop ABCD with edge length a . The resistance of the wire ABC is r and that of ADC is $2r$. The value of magnetic field at the centre of the loop assuming uniform wire is



- (1) $\frac{\sqrt{2}\mu_0 i}{3\pi a} \odot$
(2) $\frac{\sqrt{2}\mu_0 i}{3\pi a} \otimes$
(3) $\frac{\sqrt{2}\mu_0 i}{\pi a} \odot$
(4) $\frac{\sqrt{2}\mu_0 i}{\pi a} \otimes$

41. A magnet of magnetic moment M is situated with its axis along the direction of a magnetic field of strength B . The work done in rotating it by an angle of 180° will be
- (1) $-MB$
(2) $+MB$
(3) 0
(4) $+2MB$

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42. A long magnetic needle of length $2L$, magnetic moment M and pole strength m units is broken into two pieces at the middle. The magnetic moment and pole strength of each piece will be

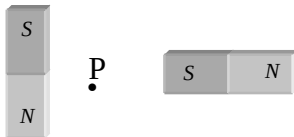
(1) $\frac{M}{2}, \frac{m}{2}$

(2) $M, \frac{m}{2}$

(3) $\frac{M}{2}, m$

(4) M, m

43. Two equal bar magnets are kept as shown in the figure. The direction of resultant magnetic field, indicated by arrow head at the point P is (approximately)



(1) $\rightarrow$

(2) $\nearrow$

(3) $\searrow$

(4) $\uparrow$

44. At a place the earth's horizontal component of magnetic field is 0.36×10^{-4} weber/m². If the angle of dip at that place is 60° , then the vertical component of earth's field at that place in weber/m² will be approximately

(1) 0.12×10^{-4}

(2) 0.24×10^{-4}

(3) 0.40×10^{-4}

(4) 0.62×10^{-4}

45. The value of the horizontal component of the earth's magnetic field and angle of dip are 1.8×10^{-5} weber/m² and 30° respectively at some place. The total intensity of earth's magnetic field at that place will be :

(1) 2.08×10^{-5} weber/m²

(2) 3.67×10^{-5} weber/m²

(3) 3.18×10^{-5} weber/m²

(4) 5.0×10^{-5} weber/m²

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46. If $\Delta_0 < P$, the correct electronic configuration for d^4 system will be
 (1) $t_{2g}^4 e_g^0$ (2) $t_{2g}^3 e_g^1$
 (3) $t_{2g}^0 e_g^4$ (4) $t_{2g}^2 e_g^2$
47. Wilkinson's catalyst is
 (1) $TiCl_4$ and $Al(C_2H_5)_3$
 (2) $[Rh(PPh_3)_3]Cl$
 (3) $TiCl_4$ and $(C_6H_5)_3P$
 (4) $Al(C_2H_5)_3$, $TiCl_4$ and $(C_6H_5)_3P$
48. The iron in haemoglobin is bound by
 (1) Hydrogen bonds
 (2) Chelation
 (3) Ionic bonds
 (4) Covalent bonds
49. Which of the following statement is true?
 (1) All amino acids contain one chiral center
 (2) All amino acids found in proteins have L configuration
 (3) In structure of valine aromatic ring is present.
 (4) All amino acids found in proteins have 1° amino group
50. For the following reaction.

$$\triangle-Br \xrightarrow[(ii) CO_2, (iii) H_3O^+]{(i) Mg, Et_2O} [A]$$

 produce [A] is :
 (1) $(\triangle-COO)_2Mg$ (2) $\triangle-COOH$
 (3) $\triangle-C(=O)-\triangle$ (4) $\triangle$
51. The oxidation number of Fe in $K_3[Fe(CN)_6]$ is
 (1) 2 (2) 3 (3) 4 (4) 6
52. ΔH_f° 298 K of methanol is given by the chemical equation :-
 (1) $CH_4(g) + \frac{1}{2}O_2(g) \longrightarrow CH_3OH(g)$
 (2) $C(\text{graphite}) + \frac{1}{2}O_2(g) + 2H_2(g) \longrightarrow CH_3OH(l)$
 (3) $C(\text{diamond}) + \frac{1}{2}O_2(g) + 2H_2(g) \longrightarrow CH_3OH(l)$
 (4) $CO(g) + 2H_2(g) \longrightarrow CH_3OH(l)$
53. Match list I with list II and select the correct answer using the codes given below the lists.
- | List I | List II |
|---------------------------|-----------------------|
| A. Nucleic acids | (i) D.N.A. |
| B. Uracil | (ii) Hormones |
| C. Thymine | (iii) Polynucleotides |
| D. double-helix structure | (iv) R.N.A. |
- (1) A-(iv), B-(iii), C-(i), D-(i)
 (2) A-(iii), B-(iv), C-(i), D-(i)
 (3) A-(iii), B-(i), C-(iv), D-(i)
 (4) none of these
54. The shapes of PCl_4^+ , PCl_4^- and $AsCl_5$ are respectively :-
 (1) square planar, tetrahedral, see-saw
 (2) tetrahedral, see-saw, trigonal bipyramidal
 (3) tetrahedral, square planar and pentagonal bipyramidal
 (4) trigonal bipyramidal, tetrahedral and square pyramidal

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- | | |
|--|---|
| <p>55. The bond angle is minimum in :-
 (1) H_2Se (2) H_2Te
 (3) H_2O (4) H_2S</p> <p>56. Choose the correct relationship for α-D-glucose (A) and β-D-glucose(B)
 (1) A and B are epimers
 (2) A and B are crystal modification
 (3) A is a pyranose sugar and B is furanose sugar
 (4) A is an aldose and B is a ketose.</p> <p>57. α-D-Glucose and β-D Glucose differ from each other due to difference in one of carbon with respect to its
 (1) Size of hemiacetal ring
 (2) Number of—OH groups
 (3) Configuration
 (4) Conformation</p> <p>58. In an octahedral structure, the pair of d-orbitals involved in sp^3d^2-hybridisation is :-
 (1) d_{x^2}, d_{xz} (2) d_{xy}, d_{yz}
 (3) $d_{x^2-y^2}, d_{z^2}$ (4) $d_{xz}, d_{x^2-y^2}$</p> <p>59. The solution of the complex $[\text{Cu}(\text{NH}_3)_4] \text{SO}_4$ in water
 (1) will give the tests of Cu^{2+} ions
 (2) will give the tests of NH_3
 (3) will give the tests of SO_4^{2-} ions
 (4) will not give the tests of any of the above species.</p> <p>60. EDTA^{4-} is a
 (1) bidentate ligand
 (2) tetradentate ligand
 (3) hexadentate ligand
 (4) flexidentate ligand.</p> | <p>61. The correct name for the complex $[\text{Cr}(\text{NH}_3)_6] [\text{Co}(\text{C}_2\text{O}_4)_3]$ is
 (1) hexaamminechromium(III) trioxalatocobalt(III)
 (2) hexaamminechromate(III) trioxalatocobalate(III)
 (3) hexaamminechromium(III) trioxalatocobaltate(III)
 (4) hexaamminechromate(III) trioxalatocobalt (III).</p> <p>62. Which one of the following statements is incorrect about Werner's theory?
 (1) Primary valency is the same thing as oxidation state
 (2) Secondary valency is the same thing as coordination number
 (3) Secondary valencies are satisfied by negative ions
 (4) Secondary valencies are directional whereas primary valencies are non-directional.</p> <p>63. The complex ions $[\text{Fe}(\text{CN})_6]^{3-}$ and $[\text{Fe}(\text{CN})_6]^{4-}$
 (1) are both octahedral and paramagnetic
 (2) are both octahedral and diamagnetic
 (3) have same structure but opposite magnetic character
 (4) have different structures but same magnetic character.</p> <p>64. The diamagnetic molecules are :-
 (1) $\text{B}_2, \text{C}_2, \text{N}_2$ (2) $\text{O}_2, \text{N}_2, \text{F}_2$
 (3) $\text{C}_2, \text{N}_2, \text{F}_2$ (4) $\text{B}_2, \text{O}_2^{2-}, \text{N}_2$</p> <p>65. The correct bond order for CO and CO^+ are respectively :-
 (1) 3, 5/2 (2) 3, 2
 (3) 3, 7/2 (4) 5/2, 3</p> |
|--|---|

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- | | |
|---|--|
| <p>66. Ferrocene is</p> <ol style="list-style-type: none"> (1) $\text{Fe}(\eta^5 - \text{C}_5\text{H}_5)_2$ (2) $\text{Fe}(\eta^2 - \text{C}_5\text{H}_5)_2$ (3) $\text{Cr}(\eta^5 - \text{C}_5\text{H}_5)_5$ (4) $\text{Os}(\eta^5 - \text{C}_5\text{H}_5)_2$ <p>67. In an octahedral crystal field, the t_{2g} orbitals are</p> <ol style="list-style-type: none"> (1) raised in energy by $0.4 \Delta_0$ (2) lowered in energy by $0.4 \Delta_0$ (3) raised in energy by $0.6 \Delta_0$ (4) lowered in energy by $0.6 \Delta_0$ <p>68. The reaction conditions leading to best yields of $\text{C}_2\text{H}_5\text{Cl}$ are :-</p> <ol style="list-style-type: none"> (1) $\text{C}_2\text{H}_6(\text{excess}) + \text{Cl}_2 \xrightarrow{\text{UV light}}$ (2) $\text{C}_2\text{H}_6 + \text{Cl}_2 \xrightarrow[\text{room temp.}]{\text{Dark}}$ (3) $\text{C}_2\text{H}_6 + \text{Cl}_2(\text{excess}) \xrightarrow{\text{UV light}}$ (4) $\text{C}_2\text{H}_6 + \text{Cl}_2 \xrightarrow{\text{UV light}}$ <p>69. Which of the following statements is incorrect?</p> <ol style="list-style-type: none"> (1) Polyethylene contains double bonds (2) The monomer used to make teflon is C_2F_4 (3) Condensation polymers are known as copolymers (4) A denatured protein could have the same primary structure as the active protein. <p>70. Melamine polymer is copolymer of</p> <ol style="list-style-type: none"> (1) Melamine and acetaldehyde (2) Melamine and formaldehyde (3) Phenol and formaldehyde (4) None of the above. | <p>71. When ethyl chloride is heated with AgCN, the main product is :</p> <ol style="list-style-type: none"> (1) ethyl cyanide (2) ethyl isocyanide (3) ethyl amine (4) ethyl nitrate <p>72. Polyacrylonitrile, characterized by the repeating unit, is made from which of the following monomer ?</p> <ol style="list-style-type: none"> (1) $\text{CH}_3\text{CH}_2\text{CN}$ (2) $\text{HOCH}_2\text{CH}_2\text{CH}_3$ (3) $\text{CH}_3\text{CH} = \text{CHCN}$ (4) $\text{CH}_2 = \text{CHCN}$ <p>73. An aldohexose (e.g., glucose) and 2-oxohexose (e.g., fructose) can be distinguished with the help of</p> <ol style="list-style-type: none"> (1) Tollen's reagent (2) Fehling's solution (3) Benedict solution (4) $\text{Br}_2 / \text{H}_2\text{O}$ <p>74. The open-chain glucose on oxidation with HIO_4 gives</p> <ol style="list-style-type: none"> (1) $5 \text{HCOOH} + \text{H}_2\text{C} = \text{O}$ (2) $4 \text{HCOOH} + 2 \text{H}_2\text{C} = \text{O}$ (3) $3 \text{HCOOH} + 3 \text{H}_2\text{C} = \text{O}$ (4) $2 \text{HCOOH} + 4 \text{H}_2\text{C} = \text{O}$ |
|---|--|

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75. Match list I with list II and select the correct answer using the codes given below the lists.

List I

List II

- A. Phenol + formaldehyde. (i) Synthetic rubber
B. Terephthalic acid + ethylene glycol (ii) Bakelite
C. Caprolactam (iii) Nylon-6
D. Butadiene + styrene (iv) terylene

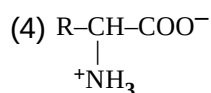
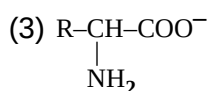
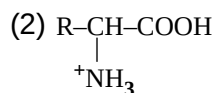
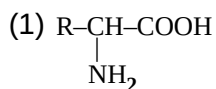
(1) A-(ii), B-(iii), C-(iv), D-(i)

(2) A-(iii), B-(i), C-(ii), D-(iv)

(3) A-(ii), B-(iv), C-(iii), D-(i)

(4) none of these

76. At the isoelectric point for an amino acid the species present are



77. Secondary structure of a protein refers to

(1) Mainly denatured proteins and structures of prosthetic groups

(2) Regular folding patterns of contiguous portions of the polypeptide chain

(3) Linear sequence of amino acid residues in the polypeptide chain

(4) None of these

78. Molich's test is done for the detection of

(1) Alkyl halide

(2) Carbohydrate

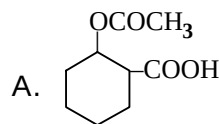
(3) Alkaloid

(4) Fat

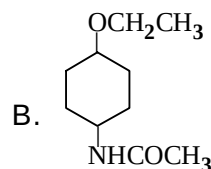
79. Match list I with list II and select correct answer using the codes given below the lists.

List I

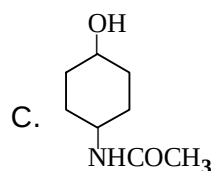
List II



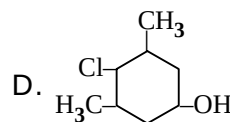
(i) Component of dettol



(ii) Bithional



(iii) Aspirin



(iv) Phenacetin

(v) Paracetamol

(1) A-(iv), B-(iii), C-(i), D-(ii)

(2) A-(iii), B-(iv), C-(ii), D-(i)

(3) A-(iii), B-(iv), C-(v), D-(i)

(4) none of these

80. A substance which can act both as an antiseptic and disinfectant is

(1) Aspirin

(2) Chloroxylenol

(3) Bithional

(4) Phenol

81. Glucose gives the silver mirror test with ammoniacal solution of silver nitrate because it contains the group

(1) Aldehyde

(2) Ester

(3) Ketone

(4) Amide

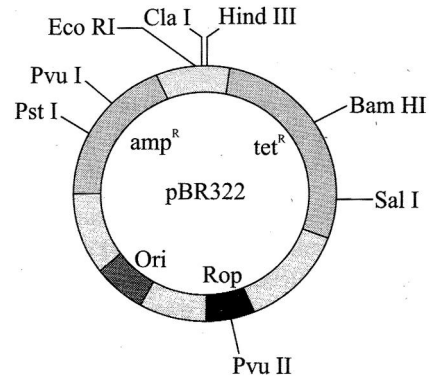
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82. The letter 'D' in carbohydrates represents
 (1) Its direct synthesis
 (2) Its dextrorotation
 (3) Its mutarotation
 (4) Its configuration
83. The enthalpy of neutralization of a strong acid by a strong base is $-57.32 \text{ kJ mol}^{-1}$. The enthalpy of formation of water is $-285.84 \text{ kJ mol}^{-1}$. The enthalpy of formation of hydroxyl ion is :-
 (1) $+228.52 \text{ kJ mol}^{-1}$
 (2) $-114.26 \text{ kJ mol}^{-1}$
 (3) $-228.52 \text{ kJ mol}^{-1}$
 (4) $+114.2 \text{ kJ mol}^{-1}$
84. The correct order of decreasing polarisability of ion is :-
 (1) $\text{Cl}^- > \text{Br}^- > \text{I}^- > \text{F}^-$
 (2) $\text{F}^- > \text{I}^- > \text{Br}^- > \text{Cl}^-$
 (3) $\text{I}^- > \text{Br}^- > \text{Cl}^- > \text{F}^-$
 (4) $\text{F}^- > \text{Cl}^- > \text{Br}^- > \text{I}^-$
85. A system absorbs 20 kJ heat and also does 10 kJ of work. The net internal energy of the system:
 (1) increases by 10 kJ
 (2) decreases by 10 kJ
 (3) increases by 30 kJ
 (4) decreases by 30 kJ
86. The standard molar heat of formation of ethane, CO_2 and $\text{H}_2\text{O}(l)$ are respectively -21.1 , -94.1 and -68.3 kcal . The standard molar heat of combustion of ethane will be :-
 (1) -372 kcal (2) -240 kcal
 (3) 162 kcal (4) 183.5 kcal
87. Ozonolysis products of an olefin are OHC-CHO and $\text{OHC-CH}_2\text{-CH}_2\text{-CHO}$, the olefin is :
 (1)  (2)  (3)  (4) 
88. The correct order for bond angles is :-
 (1) $\text{NO}_2^+ > \text{NO}_2 > \text{NO}_2^-$
 (2) $\text{NO}_2^+ > \text{NO}_2^- > \text{NO}_2$
 (3) $\text{NO}_2 > \text{NO}_2^- > \text{NO}_2^+$
 (4) $\text{NO}_2^- > \text{NO}_2 > \text{NO}_2^+$
89. Bakelite is obtained from phenol and formaldehyde. The initial reaction between the two compounds is an example of
 (1) Aromatic electrophilic substitution
 (2) Aromatic nucleophilic Substitution
 (3) Free radical reaction
 (4) Aldol reaction.
90. Match list I with list II and select the correct answer using the codes given below the lists.
- | | |
|-------------------------|---|
| <i>List I</i> (Polymer) | <i>List II</i> (Polymerizing units) |
| A. Bakelite | (i) Butadiene and styrene |
| B. Dacron | (ii) Phenol and methanal |
| C. Nylon-66 | (iii) 1,2-dihydroxyethane and dimethylterephthalate |
| D. Buna-S | (iv) Urea and methanal |
| | (v) 1, 6-hexanedioic acid and 1, 6-diamino hexane |
- (1) A-(iv), B-(iii), C-(v), D-(i)
 (2) A-(ii), B-(iii), C-(v), D-(i)
 (3) A-(v), B-(ii), C-(iii), D-(i)
 (4) none of these

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91. Gel electrophoresis is used for
- (1) Construction of recombinant DNA by joining with cloning vectors
 - (2) Isolation of DNA molecules
 - (3) Cutting of DNA into fragments
 - (4) Separation of DNA fragments according to their size
92. Bacterium commonly used in plant genetic engineering is
- (1) *Agrobacterium*
 - (2) *Corynebacterium*
 - (3) *Bacillus subtilis*
 - (4) *Salmonella typhi*
93. Which is used in gene cloning?
- (1) Lomasomes
 - (2) Mesosomes
 - (3) Plasmids
 - (4) Nucleotides
94. Which can be used as vector for transfer of DNA segment?
- (a) bacterium
 - (b) plasmid
 - (c) Plasmodium
 - (d) bacteriophage
- Options :
- (1) a, b and d
 - (2) a only
 - (3) a and c
 - (4) b and d
95. There is a restriction endonuclease called EcoRI. What does 'co' part of it stand for?
- (1) Coenzyme
 - (2) *coli*
 - (3) Colon
 - (4) Coelom
96. Agarose extracted from sea weeds finds use in
- (1) PCR
 - (2) Gel electrophoresis
 - (3) Spectrophotometry
 - (4) Tissue culture

97. In the diagram of pBR 322, which identifies components correctly?



- (1) rop—reduced osmotic pressure
 - (2) Hind III, EcoRI—selectable markers
 - (3) amp^R, tet^R—antibiotic resistance genes
 - (4) ori—original restriction enzyme
98. DNA fragments generated by the restriction endonucleases in a chemical reaction can be separated by
- (1) Electrophoresis
 - (2) Restriction mapping
 - (3) Centrifugation
 - (4) Polymerase chain reaction
99. Which of the following is not correctly matched for the organism and its cell wall degrading enzyme?
- (1) Algae—Methylase
 - (2) Fungi—Chitinase
 - (3) Bacteria—Lysozyme
 - (4) Plant cells—Cellulase
100. Eco RI cleaves the DNA strands to produce
- (1) Blunt ends
 - (2) Sticky ends
 - (3) Satellite ends
 - (4) Ori replication end
101. During the process of isolation of DNA, chilled ethanol is added to
- (1) Precipitate DNA
 - (2) Break open the cell to release DNA
 - (3) Facilitate action of restriction enzymes
 - (4) Remove proteins such as histones

102. Which of the following is a palindromic sequence?

- (1) 5'—CGTATG—3'
3'—GCATAC—5'
- (2) 5'—CGAATG—3'
3'—CGAATG—5'
- (3) 5'—GAATTC—3'
3'—CTTAAG—5'
- (4) 5'—GACTAC—3'
3'—TACGAC—5'

103. The restriction enzymes are used in genetic engineering, because

- (1) They can cut DNA at specific base sequence
- (2) They are nucleases that cut DNA at variable sites
- (3) They can degrade harmful proteins
- (4) They can join different DNA fragments

104. The introduction of T-DNA into plants involves

- (1) Altering the pH of the soil, then heat-shocking the plants
- (2) Exposing the plants to cold for a brief period
- (3) Allowing the plant roots to stand in water
- (4) Infection of the plant by *Agrobacterium tumefaciens*.

105. Which is true?

- (1) Centromere is found in animal cells which produces aster during cell division
- (2) Insulin gene is present in every body cell
- (3) Histone is formed of nucleotides
- (4) DNA has a core of ten histones

106. The EFB stands for

- (1) European Forum of biotechnology
- (2) Engineering Federation of biotechnology
- (3) European Function on biotechnology
- (4) European Federation of biotechnology

107. EcoRI cut the DNA between the bases

- (1) G and T only
- (2) G and C only
- (3) A and T only
- (4) G and A only

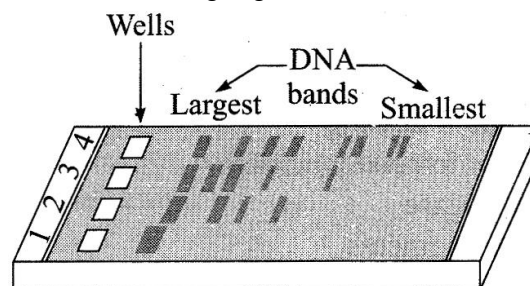
108. In pBR 322, amp^R and tet^R genes are present. When we ligate a foreign DNA at Pst-I site, then recombinant plasmids will lose the resistance to the

- (1) Ampicillin
- (2) Tetracycline
- (3) Both
- (4) None

109. An antibiotic resistance gene in a vector usually help in the selection of

- (1) Transformed cells
- (2) Non-transformed cells
- (3) Competent cells
- (4) Non-recombinant cells

110. The following figure shows



- (1) Polymerase chain reaction
- (2) Agarose gel electrophoresis
- (3) Downstream processing
- (4) Biolistics or gene gun

111. Recombinant DNA technology involves several steps in specific sequence. Find out the correct sequence.

- a. Fragmentation of DNA
- b. Culturing the host cells in a medium at large scale
- c. Ligation of DNA fragment into a vector
- d. Extraction of the desired product
- e. Isolation of DNA
- f. Isolation of desired DNA fragment
- g. Transferring the recombinant DNA into the host

- (1) e → a → f → g → c → b → d
- (2) e → f → a → c → g → b → d
- (3) a → e → c → f → g → d → b
- (4) e → a → f → c → g → b → d

- | | |
|--|--|
| <p>112. While isolating DNA from the bacteria which of the following enzyme is not used</p> <ol style="list-style-type: none"> (1) Ribonuclease (2) Deoxyribonuclease (3) Lysozyme (4) Protease <p>113. In r-DNA technique vector DNA and passenger DNA are cleaved by-</p> <ol style="list-style-type: none"> (1) Different restriction endonuclease enzyme (2) Same restriction endonuclease enzyme (3) Exonuclease (4) Any of the above <p>114. In Ti-plasmid of Agrobacterium which regions essential for transformation of plant cell:</p> <ol style="list-style-type: none"> (1) Vir-region (2) T-DNA region (3) Ori-point (4) (1) & (2) both <p>115. Which vector would be most appropriate for cloning a 1000 kbp fragment of DNA :</p> <ol style="list-style-type: none"> (1) Plasmid (2) BAC (3) YAC (4) Cosmid <p>116. Most accepted mechanism for ascent of sap is :</p> <ol style="list-style-type: none"> (1) Cytochrome pumping (2) Root pressure theory (3) Transpiration pull theory (4) Capillary theory <p>117. Which of the following plants can grow in physically dry soil:</p> <ol style="list-style-type: none"> (1) Hydrophytes (2) Xerophytes (3) Halophytes (4) Saprophytes | <p>118. Scotoactive movement of stomata is found in :</p> <ol style="list-style-type: none"> (1) Hydrophytes (2) Mesophytes (3) CAM-plants (4) Lower plants <p>119. When a cell is placed in hypertonic solution then it exhibits the plasmolysis the space between the cell wall and plasmolysed contents is filled by:</p> <ol style="list-style-type: none"> (1) Water (2) Air (3) Hypertonic solution (4) Hypotonic solution <p>120. With decrease in atmospheric pressure the rate of transpiration will</p> <ol style="list-style-type: none"> (1) Remain unaffected (2) Increase (3) Decrease slowly (4) None of above <p>121. Kidney shaped guard cells found in</p> <ol style="list-style-type: none"> (1) Mustard plant (2) Citrus plant (3) Radish plant (4) All the above <p>122. Path of ascent of sap is :</p> <ol style="list-style-type: none"> (1) Apoplastic (2) Symplastic (3) Bidirectional (4) None of above <p>123. The flow of plant sap from cut site is known as</p> <ol style="list-style-type: none"> (1) Guttation (2) Seasoning of wood (3) Exudation (4) Haemophilia <p>124. Spraying of PMA on leaves :</p> <ol style="list-style-type: none"> (1) Increase the rate of guttation (2) Increase the protien synthesis (3) Decrease the transpiration (4) None of these |
|--|--|

125. Bacteria can not survive in highly salted pickle because:
- (1) Salt inhibits reproduction in bacteria
 - (2) Bacteria do not get enough light for photosynthesis
 - (3) They become plasmolysed and consequently killed
 - (4) Pickle does not contain nutrients for bacteria
126. Latest explanation for opening of stomata is:
- (1) Photosynthesis in guard cell hypothesis
 - (2) Active K^+ transport mechanism
 - (3) Ca , ABA second messenger model
 - (4) None
127. During endosmosis water molecules diffused from water potential to
- (1) Lower, Higher
 - (2) Lower, Lower
 - (3) Higher, Lower
 - (4) Higher, Higher
128. "Transpiration is an unavoidable evil" who gave this statement :
- (1) Curtis
 - (2) Sayre
 - (3) Sachs
 - (4) Steward
129. The following figure shows:



- (1) Elution
- (2) Spooling
- (3) Biolistics
- (4) Downstream processing

130. Which is wrong statement :
- (1) Rate of transpiration increased with temperature
 - (2) Transpiration absent in succulents at day time
 - (3) Transpiration measured by potometer
 - (4) Guttation is more at noon
131. Rubber latex obtained by :
- (1) Transpiration
 - (2) Guttation
 - (3) Bleeding
 - (4) None
132. The main function of transpiration in plants is :
- (1) Ascent of sap
 - (2) Regulation of temperature
 - (3) Absorption of ions
 - (4) Water loss
133. Which phenomenon from below is given only for closing of stomata :
- (1) Sugar - starch interchange theory
 - (2) Malate - switch hypothesis
 - (3) Calcium ABA second messenger model
 - (4) K^+ ion theory
134. Denitrification means :
- (1) Conversion of atmospheric nitrogen to ammonia
 - (2) Conversion of atmospheric nitrogen to nitrate
 - (3) Conversion of atmospheric nitrogen to nitrite
 - (4) Conversion of nitrate to nitrogen
135. Thermal cycle is used in :-
- (1) Radioactivation
 - (2) Chemical reaction
 - (3) PCR
 - (4) Enzyme catalyzed reaction

136. Identify the given figure.



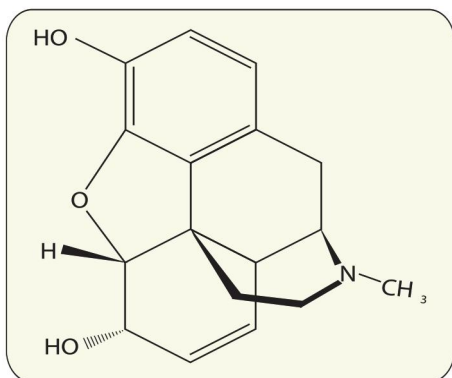
select correct statements.

- (I) Opiates are obtained from this plant.
 - (II) It is member of family cannabaceae
 - (III) It is commonly known as poppy plant.
- (1) I and III (2) I and II
(3) II and III (4) I, II and III

137. Smack is -

- (1) Obtained by acetylation of morphine
- (2) White and odourless
- (3) Bitter crystalline compound
- (4) All of these

138. Identify the compound on the basis of given chemical structure and select its main therapeutic use-



- (1) To increase the activity of cardiovascular system
- (2) As an analgesics
- (3) As an antipyretic
- (4) All of these

139. Which is not true for cocaine ?

- (1) Commonly called coke
- (2) Obtained from *Erythroxylum coca*
- (3) Produce sense of euphoria
- (4) None of these

140. Cannabinoid receptors are present principally in-

- (1) Brain (2) Liver
- (3) Heart (4) Lungs

141. Which of the following has stimulatory effect on central nervous system.

- (1) Morphine (2) Smack
- (3) Codein (4) Crack

142. How many of the following are direct or indirect effects of nicotine ?

- (I) Release of adrenaline and nor-adrenaline in to blood circulation.
 - (II) Increased blood pressure
 - (III) Increased heart beat rate.
- (1) 3 (2) 2 (3) 1 (4) 0

143. Smoking is associated with increased incidence of cancer of -

- (1) Lungs (2) Urinary bladder
- (3) Throat (4) All of these

144. How many of the following are cannabinoids? Marijuana ; Hashish ; LSD ; Ganja

- (1) 2 (2) 1 (3) 3 (4) 4

145. Select the organs which are affected by smoking-

- (A) Lungs; (B) Heart ;
 - (C) Arteries; (D) Stomach
- (1) A and B only
(2) B and C only
(3) A, B and C only
(4) A, B, C and D

146. Which of the following disorder is associated with alcoholism ?

- (1) Wernicke- Korsakoff syndrome
- (2) Bantis Syndrome
- (3) Edward Syndrome
- (4) Cushing's syndrome

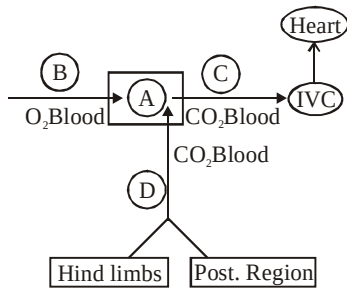
147. Select the disorder which can be caused due to alcoholism-

- (1) Neuritis
- (2) Liver Cirrhosis
- (3) Gastritis
- (4) More than options are correct

148. Select correct statements for barbiturates.
- (I) Used as medicines to cope with mental illness like depression and insomnia.
 (II) These are synthetic drugs derived from barbituric acid.
 (III) Have stimulatory effect on CNS.
- (1) I and II (2) I and III
 (3) II and III (4) All I, II and III
149. Select the side effects of anabolic steroids which can occur in both males and females-
- (I) Increased aggressiveness
 (II) Mood swings
 (III) Depression
- (1) I and II (2) II and III
 (3) I, II and III (4) Only III
150. Which of the following has teratogenic effect ?
- (1) Alcohol (2) Tobacco
 (3) Thalidomide (4) All of these
151. Select the statement(s) about adolescence :-
- (1) The period between 12-18 years is adolescence period
 (2) Adolescence is a bridge linking childhood and adulthood.
 (3) It is a very vulnerable phase of mental and psychological development of an individual
 (4) All of these
152. Drugs which include dreamy state of consciousness are called :
- (1) Sedatives (2) Barbiturates
 (3) Stimulants (4) Hallucinogens
153. Which one of the following statement is false :
- (1) Psychoactive drugs alter the activity of brain
 (2) Hallucinogens change person's perceptions
 (3) Adolescence is characterized by accelerated physical growth, development of reproductive organs and changes in functioning of neuroendocrine system
 (4) Heroin is a stimulant
154. The drug which stimulates the nervous system and makes it more alert and active is called :
- (1) Sedative
 (2) Opiate narcotic
 (3) Stimulant
 (4) Hallucinogen
155. In drunk person, the part of brain to be affected first is :-
- (1) Cerebellum
 (2) Pons varolli
 (3) Medula oblongata
 (4) Cerebrum
156. Antisleep pills are made of :
- (1) Barbiturates
 (2) Benzodiazepines
 (3) Amphetamines
 (4) Both (1) and (2)
157. Select the correct statements from the ones given below :
- (1) Cocaine is given to patients after surgery as it stimulates recovery
 (2) Barbiturates when given to criminals make them tell the truth
 (3) Morphine is often given to persons who have undergone surgery as pain killer
 (4) Chewing of tobacco lowers the blood pressure and heart rate
158. Which is incorrect sentence :
- (1) Nicotine stimulates the adrenal medulla and it causes release of adrenaline & nor adrenaline hormones in the blood and blood pressure and heart rate increases
 (2) Smoking is also associated with increased incidence of coronary heart disease and gastric ulcer
 (3) Opioids receptors are present in CNS and gastro intestinal tract of our body
 (4) Morphine is used to treat mental illnesses like depression and insomnia

159. Which measure is not included in prevention and control of alcohol & drugs abuse among adolescents:
- (1) Avoid undue peer pressure
 - (2) Looking for danger signs
 - (3) Seeking help from parents and peer
 - (4) Not take help from medical professional
160. is the tendency of the body to manifest a characteristic and unpleasant syndrome :
- (1) Addiction, acquire
 - (2) Deaddiction, acquire
 - (3) Dependence, withdrawal
 - (4) Dependence, Down's
161. A person is taking drugs for chronic time then :
- (1) the tolerance level of the receptors of person will increased and it not respond to small dose of drugs
 - (2) the numbers of receptors in body are increased and person will respond to only larger dose of drugs
 - (3) the tolerance level of the receptors remain same but responding power of receptors decreased
 - (4) Both (2) & (3)
162. Majority of opioids, cannabinoids and coca-alkaloids are obtained from :
- (1) Algae
 - (2) Fungi
 - (3) Plants
 - (4) Animals
163. Heroin is obtained by :
- (1) Accetylation of LSD
 - (2) Accetylation of amphetamines
 - (3) Accetylation of cannabinoids
 - (4) None of them is correct option
164. Which of the following does not have the property of hallucination :
- (1) LSD
 - (2) Atropa belladonna
 - (3) Cocains
 - (4) Morphine
165. A plant , native of south america is a source of a drug of addiction which is CNS stimulant . This drug inhibits transport of
- (1) Acetylcholine
 - (2) Dopamine
 - (3) Serotonin
 - (4) Adrenaline
166. In which of the following possibility of erythroblastosis foetalis is maximum ?
- (1) Father Rh⁺, Mother Rh⁺, Foetus Rh⁻
 - (2) Father Rh⁻, Mother Rh⁻, Foetus Rh⁻
 - (3) Father Rh⁺, Mother Rh⁻, Foetus Rh⁺
 - (4) Father Rh⁺, Mother Rh⁻, Foetus Rh⁻
167. Which is correct with respect to all leucocytes :-
- (1) Neutrophils - 2-3 lobed nucleus, 20% - 25%
 - (2) Monocytes - bilobed nucleus, 5% - 8%
 - (3) Basophils - 2-3 lobed nucleus, 0.5% - 1%
 - (4) Acidophils - 3-7 lobed nucleus, 2% - 3%
168. Right atrioventricular apperture is guarded by :-
- (1) Mitral valve
 - (2) Tricuspid valve
 - (3) Bicuspid valve
 - (4) Semilunar valve
169. If you are a cardiac surgeon and you observed that 85% blockage occured in coronary artery of patient through diagnosis. So at the time of CABG technique which artery or vein will choose for bypass channel :-
- (1) Brachial artery
 - (2) Segment of coronary artery
 - (3) Segment of saphenous vein
 - (4) Iliac vein
170. Consider the following statements (a–d) and select the option which includes all the correct ones only:-
- (a) At the time of ventricular systole AVV are closed and SLV are opened.
 - (b) At the time of ventricular diastole AVV are opened and SLV are closed.
 - (c) At the time of joint diastole AVV are opened and SLV are closed.
 - (d) At the time of atrial systole AVV are closed and SLV are opened.
- Options :-
- (1) Statements a, b and c
 - (2) Statements a and b only
 - (3) Statements c and d only
 - (4) Statements b, c and d

171. Given below a sketchy diagram of blood circulation in specific region of frog.



In above figure what are the A, B, C and D and select the correct option :-

- (1) A = Kidney,
B = Renal vein,
C = Renal artery,
D = Renal portal vein
- (2) A = Liver,
B = Hepatic artery,
C = Hepatic vein,
D = Hepatic portal vein
- (3) A = Kidney,
B = Renal artery,
C = Renal vein,
D = Renal portal vein
- (4) A = Kidney,
B = Renal artery,
C = Renal portal vein,
D = Renal vein

172. What will happen if the neuromuscular conduction pathway in heart is broken between S.A. node and A.V. node :-

- (1) Auricles will contract due to impulse generated by S.A. node, while ventricles will not show any activity
- (2) No contraction or heart beat will be seen in heart
- (3) Auricles will contract at the rate set by the S.A. node while ventricles will contract at the rate of impulse generated by the A.V. node
- (4) There will be no effect on the activity of heart

173. In a study of four blood vessels A, B, C & D in man, vessel 'A' carried pure blood away from heart. Vessel 'B' carried impure blood towards heart. Vessel 'C' carried pure blood into the heart musculature while vessel 'D' carried impure blood away from heart. The correct names of the four vessels can be :-

- (1) Coronary artery–A,
Pulmonary vein–B,
Pulmonary Artery–C,
SVC–D
- (2) Pulmonary vein–A,
Pulmonary Artery–B,
Coronary vein–C,
Coronary Artery–D
- (3) Systemic Aorta–A,
Vena cava–B,
Coronary artery–C,
Pulmonary vein–D
- (4) Systemic Aorta–A,
Vena cava–B,
Coronary Artery–C,
Pulmonary Artery–D

174. In a human cardiac cycle which of the following statement is false :-

- (1) The time taken for one cardiac cycle decreases in Tachycardia
- (2) The time interval for joint Diastole in heart is about 0.4 seconds
- (3) The time interval for joint systole in heart is about 0.4 second
- (4) Left ventricle takes 0.5 seconds to complete one diastole

175. Match the blood vessels of human heart listed under column-I with the functions given under column-II: Choose the answer which gives the correct combination of the alphabets of the two columns.

	Column-I (Blood vessel)		Column-II (Function)
(A)	Superior vena cava	p	Carries deoxygenated blood to lungs
(B)	Inferior vena cava	q	Carries oxygenated blood to lungs
(C)	Pulmonary artery	r	Brings deoxygenated blood from lower parts of the body to the right atrium
(D)	Pulmonary vein	s	Brings oxygenated blood to the left atrium
		t	Brings deoxygenated blood from upper parts of the body into the right atrium

- (1) A = t, B = p, C = r, D = q
 (2) A = t, B = r, C = p, D = s
 (3) A = s, B = t, C = r, D = p
 (4) A = t, B = p, C = q, D = r

176. The course of blood from heart to the lungs and back to the heart is called :-

- (1) Systemic circulation
 (2) Pulmonary Circulation
 (3) Blood circulation
 (4) Single circuit circulation

177. Mark the correct statement/statements :

- (a) Crocodiles have 4-chambered heart
 (b) Atrio-ventricular septa are provided openings which are guarded by cuspid-valves.
 (c) Beat volume is 70 ml of each ventricle.

- (1) a, b (2) b, c
 (3) a, c (4) a, b, c

178. If following data belongs to an adult male

$$\text{Cardiac index} = \frac{\text{Cardiac output}}{\text{Total body surface area}}$$

- Cardiac index – 4 litre/mt./m²
 Heart rate – 68/minute
 End systolic volume – 50 ml.
 Total body surface area – 1.7 m²

Then choose the correct set of option.

- (1) Cardiac output = 5 litre/minute., stroke volume = 70 ml. ; EDV = 120 ml.
 (2) C.O. = 5 litre/minute. ; SV = 100 ml.; EDV = 50 ml.
 (3) C.O. = 6.8 litre/minute., SV = 50 ml; EDV = 120 ml.
 (4) C.O. = 6.8 litre/minute. ; SV = 100ml.; EDV = 150 ml.

179. If six liters of deoxygenated blood enters the lungs via pulmonary artery in one minute time then the amount of oxygenated blood which will exit the lungs via pulmonary veins in a minute time will be:-

- (1) 6 litre
 (2) < 6 litre
 (3) > 6 litre
 (4) may be more or less than 6 litre

180. (a) SAN is responsible for initiating and maintaining the rhythmic contractile activity of heart.

- (b) Body has ability to alter the stroke volume as well as heart rate and there by the cardiac output.

- (c) Cardiac output of an athlete will be much lower than that of an ordinary man.

- (d) Normal activities of the heart are regulated intrinsically.

Mark the correct sequence of true and false

- (1) TFFT
 (2) FTTF
 (3) TTFT
 (4) TFTF